STORMDOM

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Hurricane ODALYS Tracker

Hurricane POLO Tracker

Tropical Cyclone 01B Tracker

Tropical Storm SURIGAE Tracker

Tropical Storm NOLO Tracker

Tropical Storm FAY Tracker

Tropical Cyclones FAY Tracker

See my new article: *How I Met Hurricane Camille*

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the stronger storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots, 63 km/h). As the system strengthens, it becomes a tropical storm with wind speeds from 39 to 73 mph (34-63 knots, 63-118 km/h). Storms are named in the Atlantic, East Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots, 119 km/h), the storm becomes: In the Atlantic Ocean, East Pacific, central Pacific (part of the International Dateline), and Northwest Pacific (west of the International Dateline) it is a **Hurricane**. In the west part of the International Dateline) it is a **Typhoon**; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) it is a **Severe Tropical Cyclone**; in the North Indian Ocean it is **Severe Cyclonic Storm**; and in the Southwest Indian Ocean (west of 90°E) it is **Tropical Cyclone**.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph, 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph, 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged. Boats torn away. Some trees uprooted. Low-lying areas flooded.

Category 3 – 96-112 knots (111-129 mph, 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Sewers surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph, 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding occurs. The area is usually uninhabitable for weeks or months. Large fresh-water shelters must be built. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph, 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

SAFFIR-SIMPSON SCALE

Category	Knots	MPH	KM/H	Damage
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	150+	150+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly increasing ocean depth at the sea/land interface. Conversely, storm surge will be lessened in shallow-water environments, such as the Florida Keys, where the ocean bottom gradually increases depth.

Storm surge is caused by the hurricane's powerful winds pushing water in front of the storm. The stronger the storm, the greater the storm surge. Storm surge is usually highest near the right front quadrant of the storm in the Northern Hemisphere and the left front quadrant in the Southern Hemisphere. This is where the hurricane's winds are blowing in the same direction as the storm's forward motion. Because storm surge is caused by the powerful winds pushing water in front of the storm, it must also occur in the opposite direction from the storm's forward motion. Therefore, storm surge is usually highest near the right front quadrant of the storm in the Northern Hemisphere and the left front quadrant in the Southern Hemisphere. This is where the hurricane's winds are blowing in the same direction as the storm's forward motion. Because storm surge is caused by the powerful winds pushing water in front of the storm, it must also occur in the opposite direction from the storm's forward motion. Therefore, storm surge is usually lowest near the right front quadrant of the storm in the Northern Hemisphere and the left front quadrant in the Southern Hemisphere. This is where the hurricane's winds are blowing in the opposite direction from the storm's forward motion.

Tropical Cyclone Formation

By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure cyclones develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even sometimes, due westward, between 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a "convergence", when lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives in a substantively wind in the axis of the tropical wave approaches. Gusts up to 25 mph (40 km/h) from the northeast can be quite noticeable even from a distance of several hundred miles. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.

When the wave passes over warmer waters (SSTs), convection and resulting rainfall are enhanced. This low-pressure rainfall is concomitant with falling surface pressures. By the time the wave progresses far enough westward, usually to 100 mb, it is likely that the northeast wind has closed off to a southwest wind on the backside of the wave. The development of the tropical wave completes the closure on the northern side of a broad low-level circulation center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. Some of the factors that can prevent a tropical depression from tropical development are: excessive upper-level winds, dry air in the middle troposphere, or the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives in a substantively wind in the axis of the tropical wave approaches. Gusts up to 25 mph (40 km/h) from the northeast can be quite noticeable even from a distance of several hundred miles. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.

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Fire formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The overall degree to make in appearance, usually on the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and stronger, the center contracts from about 200 miles across to roughly 50 miles at this stage. An increase in rotational velocity accompanies the smaller, more defined center. The inflow is spiralling in ever faster as it is evacuated up through the developing eyewall and out by the high altitude outflow layer. The eye begins to appear as a ragged hole in the cloud pattern. The eye is typically a region of sinking air that not only helps clear the eye, but can also produce a feature known as the "moat", which is an area of relatively weak convection outside the core of the tropical cyclone. The circulation of the moat can make the storm's core look like a doughnut. The eye can form in diameter anywhere from 10 to 40 miles across.

Hurricanes exhibit a great deal of durability provided that the upper level support remains and the southerly (Northern Hemisphere), or tropical, inflow is present. Here are some examples: In 1998, Hurricane Mitch developed into a supersevere Category 5 hurricane. Nevertheless weakening wind shear allowed the center moved to a position 200 miles north of Honduras, spanning off the southern inflow and producing a tremendous amount of rainfall. Mitch's slow motion combined with the extreme rainfall rates caused widespread flooding and mudslides. Elena and Iverted, where the proximity of the eye to land is not necessarily a weakening factor, although the extensive shelf water can sometimes be an inhibiting factor, if a slow-moving storm generates enough upwelling. A recent Southern Hemisphere example of this is Severe Tropical Cyclone Monica, which developed into a "super cyclone" just north of Australia. The northerly (equatorward) inflow over the Arafura Sea fueled this system.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and Landfall.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the warm action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area equal to 300 miles across. The result is usually that it becomes possible for a hurricane to weaken or even landfall for dissipates. In 2004, Hurricane Frances stalled off the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 1 hurricane, instead of the earlier expected Category 3.

Entraining dry air. Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as the storm interacts with frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains enough, the result can be dissipation.

Moving over cool water. Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begins to weaken. This is because evaporation, particularly in the East Pacific, is the driving force behind a tropical cyclone's circulation. Without the energy source of warm water, the storm will weaken and dissipate.

Exposure to upper-level westerlies. Frequently, the edge of high pressure that keeps the hurricane heading toward the west erodes. In the Atlantic Ocean, this usually results in a strong upper-level trough moving down and impinging on the northern edge of the hurricane's upper level support structure. Poleward outflow is restricted. A hurricane can still maintain itself when outflow is restricted in one quadrant. It outflow becomes too restricted, resulting shear can disrupt the inner core.

After landfall. Tropical cyclones cannot survive over land, since their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project tide-threatening hurricane force winds over two hundred miles inland. As the storm progressed inland, it can change a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States—which caused unprecedented rainfall and giant mesh cyclones in Virginia. Moreover, the Mississippi Valley and eventually Central Illinois suffered severe flooding as the remnants of Camille moved up the Mississippi River. Hurricane Agnes in 1972 produced similar, though less dramatic, results. Agnes developed into a hurricane while moving over land, but it quickly evolved into a frontal cyclone that produced widespread flooding as it moved from the Atlantic Ocean to the Northeastern United States and inland flooding. The right-hand side of a hurricane's circulation is the Northern Hemisphere, left-hand in the Southern Hemisphere) is where the strongest winds are found. The left-hand side of the hurricane is where the storm's forward motion is added to the counter-clockwise pinch of the storm. This is the right-hand side of the hurricane where the strong on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.

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